

Motivation

Landslides cause **severe casualties and infrastructure damage** across Sri Lanka's central highlands. Post-event mapping is critical for emergency response and risk reduction, yet conventional visual interpretation of drone data is time-consuming and operator-dependent especially when multiple simultaneous events occur after a single rainfall episode.

Deep learning segmentation offers a path toward semi-automated mapping. Its performance, however, depends critically on which input data channels are available. This study asks:

Does adding a single LiDAR-derived terrain feature elevation, hillshade, or slope to high-resolution RGB drone imagery improve U-Net landslide segmentation compared to RGB alone, and if so, which feature matters most?

Study Area & Dataset

12 post-landslide drone survey sites across Sri Lanka's central highlands:

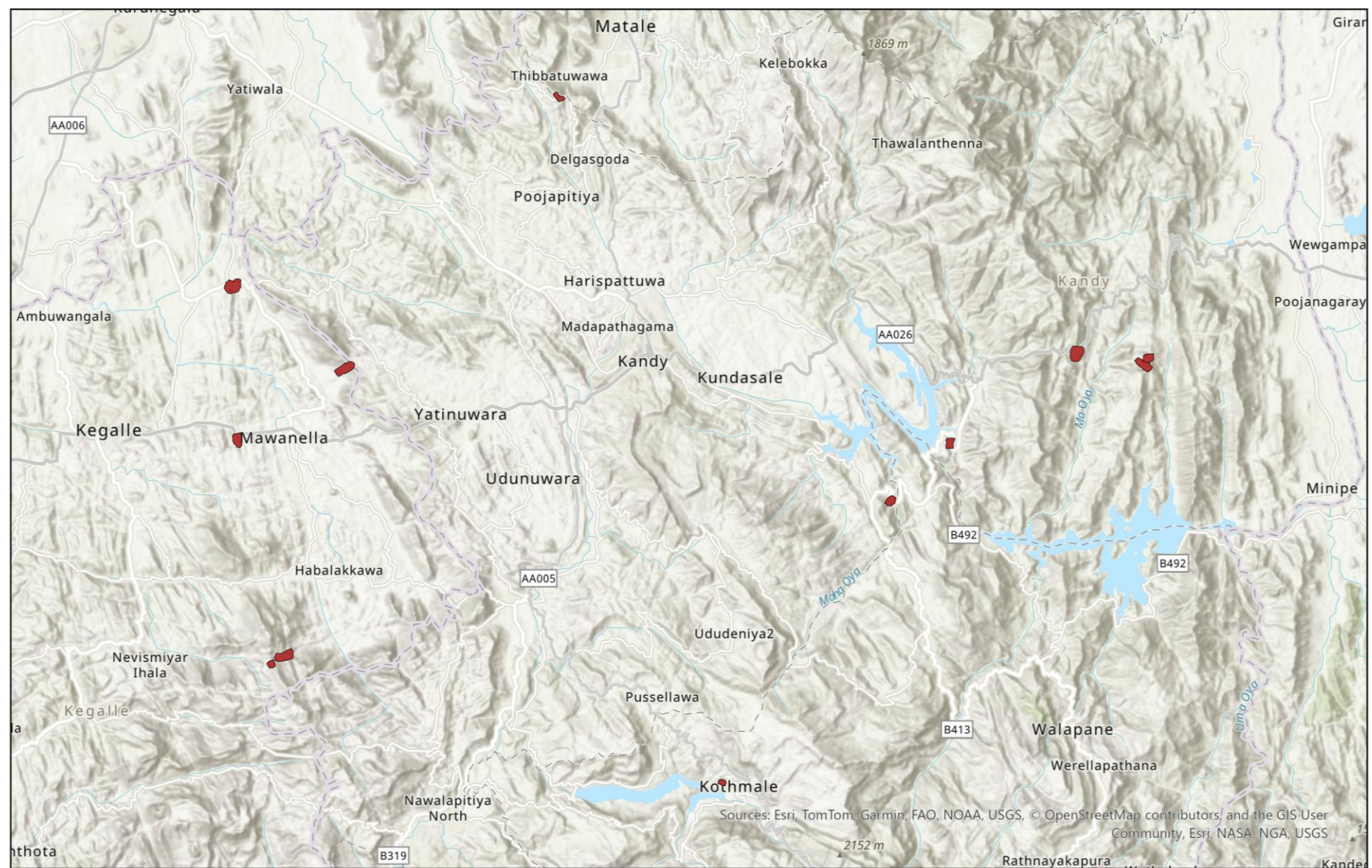


Figure 1: Overview of the 12 drone landslide survey sites in Sri Lanka

Input data:

- LiDAR point cloud → ground-classified DTM at 0.1 m grid
- RGB orthomosaic at < 5 cm GSD
- Derived layers: Hillshade (azimuth 315°, elevation 45°) and Slope

Coordinate Reference System: EPSG:5235 (Sri Lanka Datum 1999)

Train / Validation / Test split:

- Sites 01–10 → 80%: Training 20%
- 13,449 training / 3,957 validation (sites 01–10)
- 3,690 held-out test (sites 11–12)

Tiling & class balance:

- 512 × 512 px tiles, every site fully tiled
- 21,096 tiles across all 12 sites.
- Landslide pixels ≈ 8.2% of tiles

Labelling:

Binary masks (landslide / background) manually delineated per site from orthomosaic and DSM visual interpretation.

Input Configurations

- RGB**
3 channels: red, green, blue from Orthomosaic
- RGB + DTM**
4 channels: + digital terrain model at 0.1 m
- RGB + Hillshade**
4 channels: + terrain illumination (sun azimuth 315°, elevation 45°)
- RGB + Slope**
4 channels: + slope gradient (degrees)

U-Net Architecture & Training

Deep Learning Architecture

Model: U-Net based semantic segmentation framework, with a ResNet50 encoder pretrained on ImageNet

Architecture

Encoder decoder network with skip connections

Encoder:

ResNet50 (ImageNet-pretrained): 7×7 stem convolution, then 4 stages of bottleneck residual blocks (3×3 conv, Batch Normalization, ReLU) with strided downsampling between stages. Frozen for the first 4 epochs (decoder-only training), then fully unfrozen and fine-tuned at a 10x lower learning rate.

Bottleneck:

Deep feature representation with 2048 channels

Decoder:

Nearest-neighbor upsampling at each stage, followed by feature fusion with the matching encoder skip connection and two 3×3 conv, Batch Normalization, ReLU blocks

Output Layer:

1×1 Convolution → Sigmoid activation
Generates **binary landslide probability map**

Training hyperparameters:

Parameter	Configuration
Learning Rate	1×10^{-4} (decoder), 1×10^{-5} (encoder, after unfreeze)
Batch Size	8
Training Epochs	up to 100, early stopping (patience 12); best checkpoint at epoch 9 -16, runs halted 21–28 epochs in
Loss Function	BCE + Dice, pos_weight computed from data and capped at 10
Input Tile Size	512 × 512 pixels

Table 1: Training hyperparameters

Terrain feature derivation:

- DTM: ground-classified LiDAR rasterised at 0.1 m
- Hillshade: sun azimuth 315°, elevation 45° (single illumination)
- Slope: first-order gradient magnitude from DTM (degrees)

Training & Validation Loss Curves

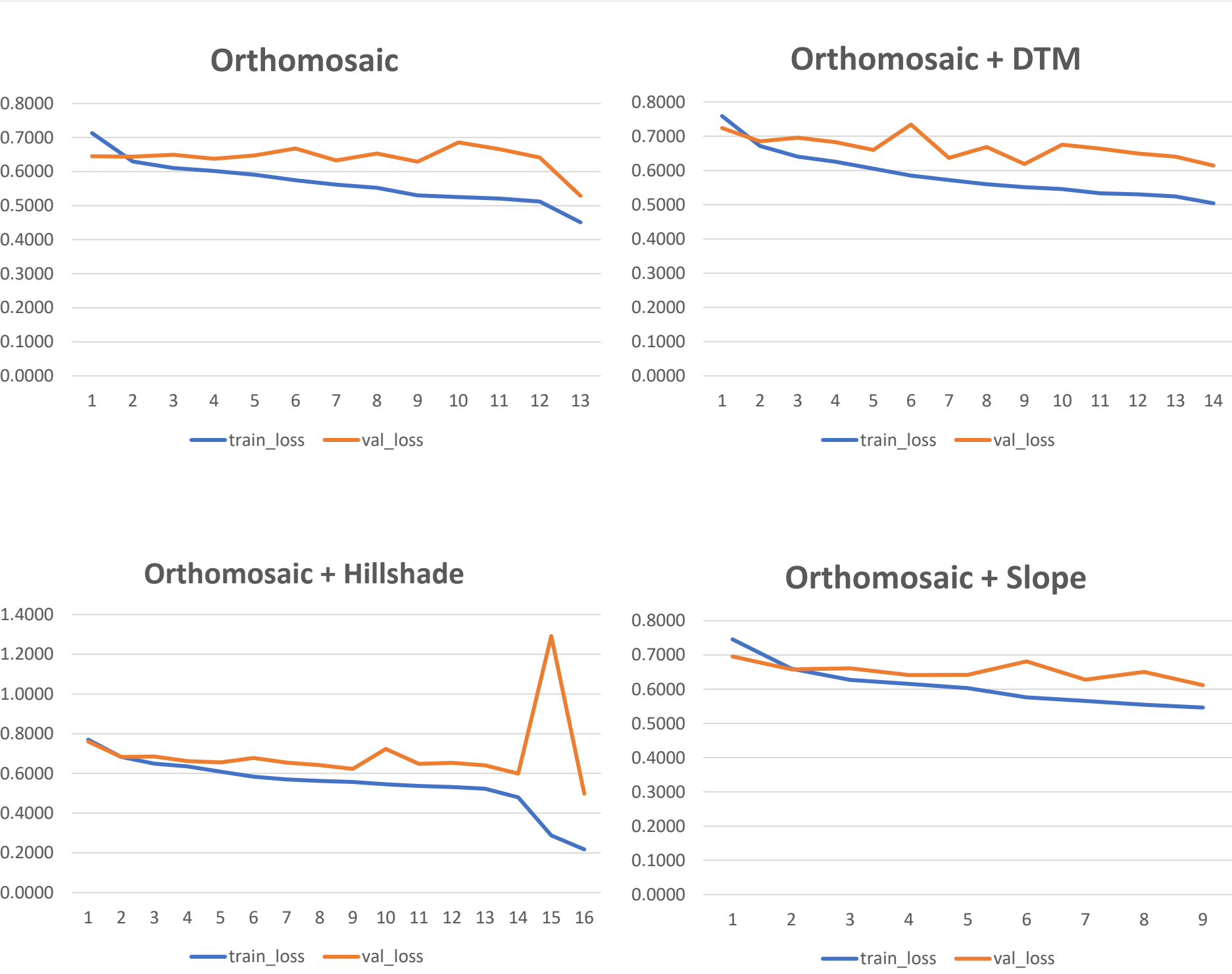


Figure 2: Train / validation loss for all four configurations

Results

Configuration	Acc.	Prec.	Recall	F1
Orthomosaic	0.972	0.440	0.949	0.599
Orthomosaic + DTM	0.985	0.593	0.825	0.690
Site 11 (Uthuwana Kanda)	0.987	0.663	0.911	0.768
Site 12 (Victoria Reservoir)	0.982	0.522	0.738	0.612
Orthomosaic + Hillshade	0.970	0.420	0.932	0.577
Orthomosaic + Slope	0.972	0.439	0.911	0.589

Table 2: Overall metrics

Results & Conclusion

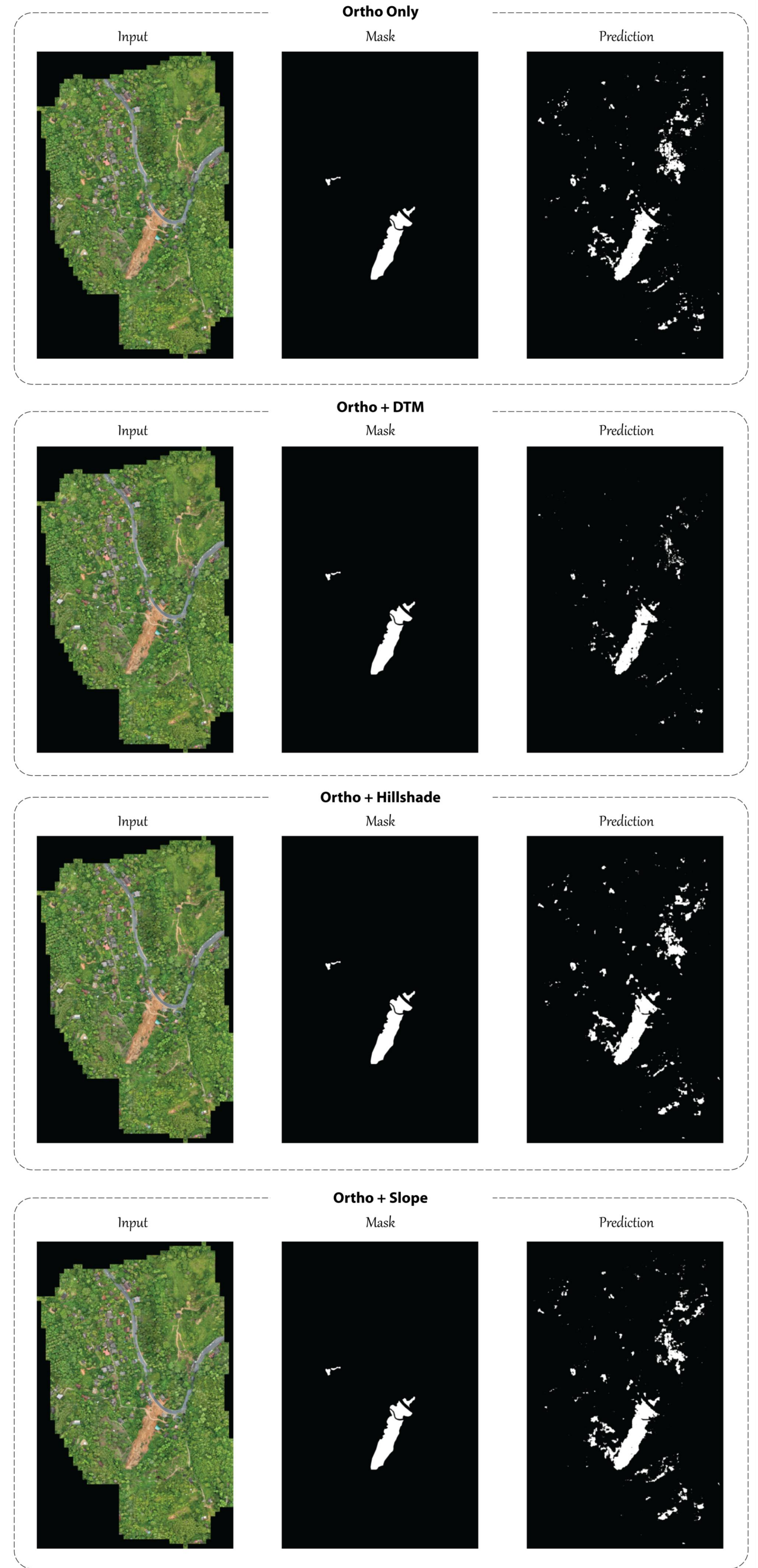


Figure 4: Site 11 Input / Mask and Prediction

- All four configurations (RGB, RGB+DTM, RGB+Hillshade, RGB+Slope) produced functional detectors, with recall consistently high (0.83–0.95) across the board.

- RGB + DTM performed best overall** (F1 = 0.690, Precision = 0.593), improving on the RGB-only baseline (F1 = 0.599) mainly by cutting false positives.

- RGB + Hillshade was the weakest configuration** (F1 = 0.577) below the RGB-only baseline, despite being derived from the same DTM.

- On Site 11, the model predicted some genuinely landslide-affected areas that the ground-truth mask doesn't fully cover.

- Because fresh landslide scars share the same chaotic, exposed-ground texture as other disturbed surfaces, the model sometimes **confuses visually similar non-landslide terrain for landslide**.

- This is most visible on Site 12, where a **gravel road was misclassified as landslide-affected ground**. However, the same kind of gravel road was correctly **excluded from the predicted mask on Site 11**.

- The **predicted mask tracks the core landslide boundary cleanly**, showing the model has learned a real landslide signature; the edges and surrounding disturbed terrain are where it still needs work.

Note: Site 12 is included in the GitHub repository due to space restriction.